

Automatic Biomedical Brain Tumor Segmentationfor DiseaseDiagnosis using Artificial Intelligence

A.Sumana-245323753003
R.Poorna Chandu-245323753051
P.Bobby Aniket-245323753047

Under the Guidance of
Dr. K Vinuthna Reddy, Associate Professor

INTRODUCTION

Project Overview

Problem

Brain tumor detection mainly relies on manual MRI analysis by radiologists, which is time-consuming and prone to errors. Small tumor regions may be overlooked, delaying diagnosis and treatment.

Solution

We developed an AI-based brain tumor segmentation system that automatically analyzes MRI scans using an enhanced U-Net deep learning model to accurately identify and segment tumor regions.

Why It Matters

The system enables early and accurate tumor detection, reduces doctors' workload, speeds up diagnosis, and supports better treatment planning, ultimately improving patient outcomes.

INTRODUCTION

Project Objective

Main Goal

To develop an AI-powered brain tumor segmentation system that automatically analyzes MRI images and accurately identifies tumor regions using deep learning, enabling faster and more reliable diagnosis.

Supporting Objectives

- Develop an automated system for accurate brain tumor segmentation from MRI scans.
- Preprocess MRI images to improve image quality and model performance.
- Implement an enhanced U-Net architecture for precise tumor detection and segmentation.
- Reduce manual effort and assist radiologists in making faster clinical decisions.
- Improve diagnosis accuracy to support timely treatment planning and better patient outcomes.

System Analysis

Existing System

- Brain tumors are detected through manual MRI image analysis.
- Diagnosis depends heavily on radiologists' expertise.
- The process is time-consuming and labor-intensive.
- Human errors may occur due to fatigue or complex MRI scans.
- Limited automation increases the workload of healthcare professionals.

Proposed System

- Uses an AI-based automated brain tumor segmentation system.
- MRI images are preprocessed to improve image quality.
- Enhanced Modified U-Net model accurately identifies tumor regions.
- Generates segmented tumor images with minimal human intervention.
- Provides fast, reliable, and consistent diagnostic support.

Advantages

- Faster MRI analysis
- Accurate tumor segmentation
- Precise boundary detection
- Supports surgery planning
- Assists radiation therapy
- Tumor growth monitoring
- Consistent AI-assisted results

TARGET AUDIENCE

Who We Designed For

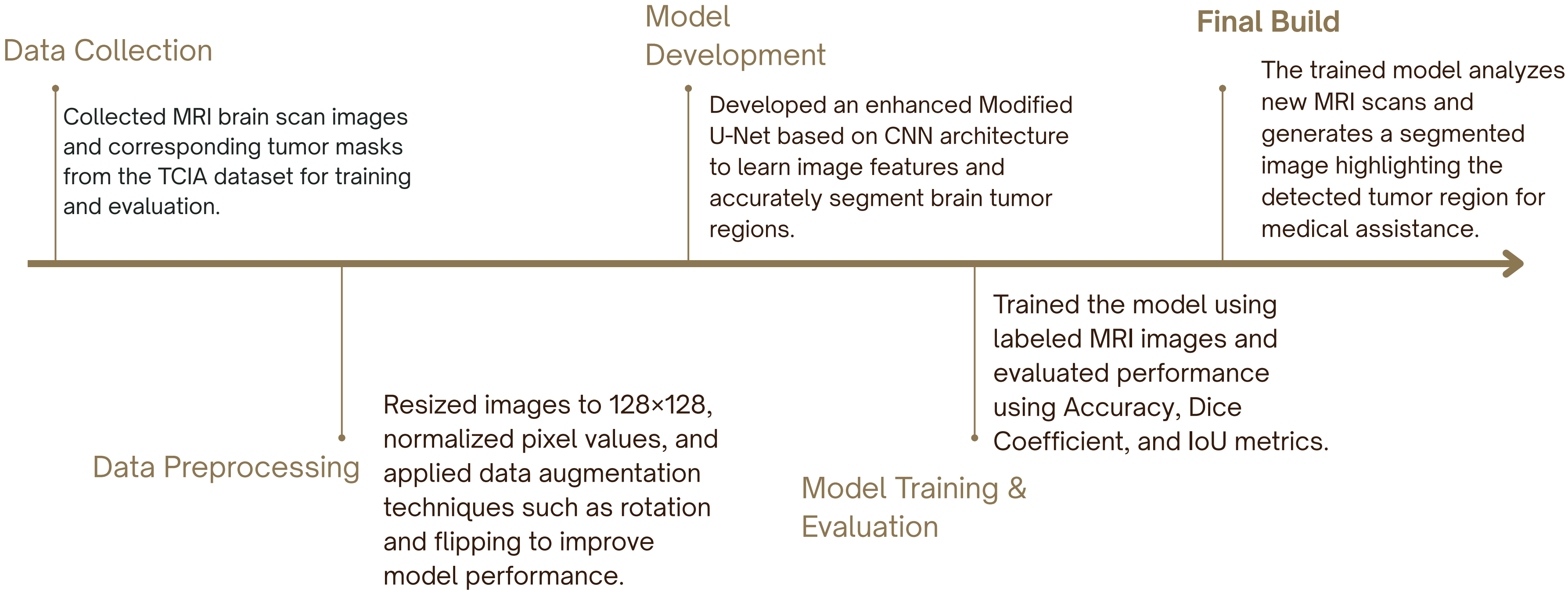
Target User

Our AI-based brain tumor segmentation system is designed for radiologists, neurologists, hospitals, diagnostic centers, and medical researchers who analyze MRI scans for brain tumor detection and treatment planning.

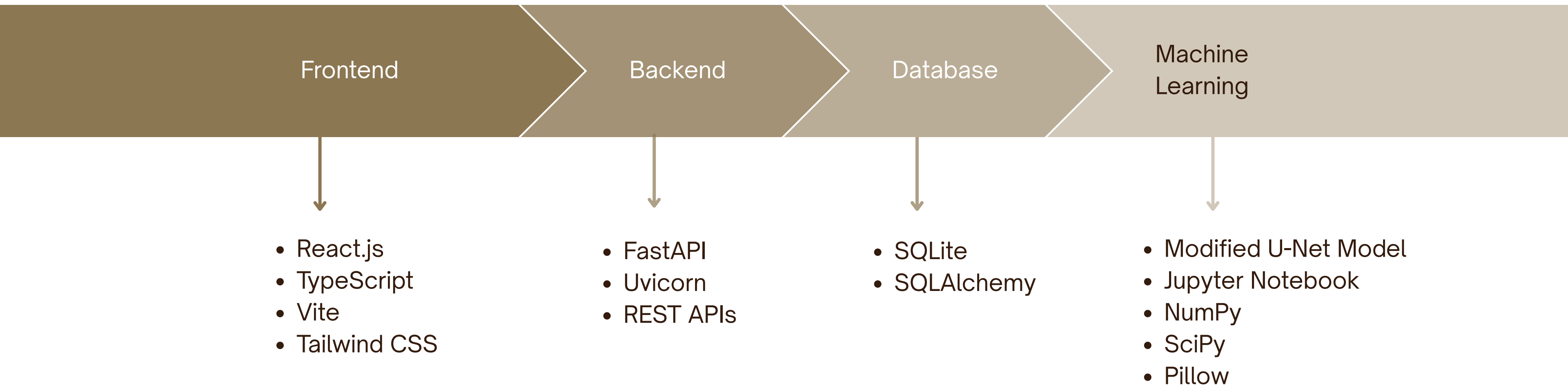
Their Challenges

- Manual MRI analysis is time-consuming and requires expert knowledge.
- Small or complex tumors can be difficult to detect accurately.
- Heavy workload may lead to delayed diagnosis and human errors.
- Accurate tumor segmentation is essential for effective treatment planning.

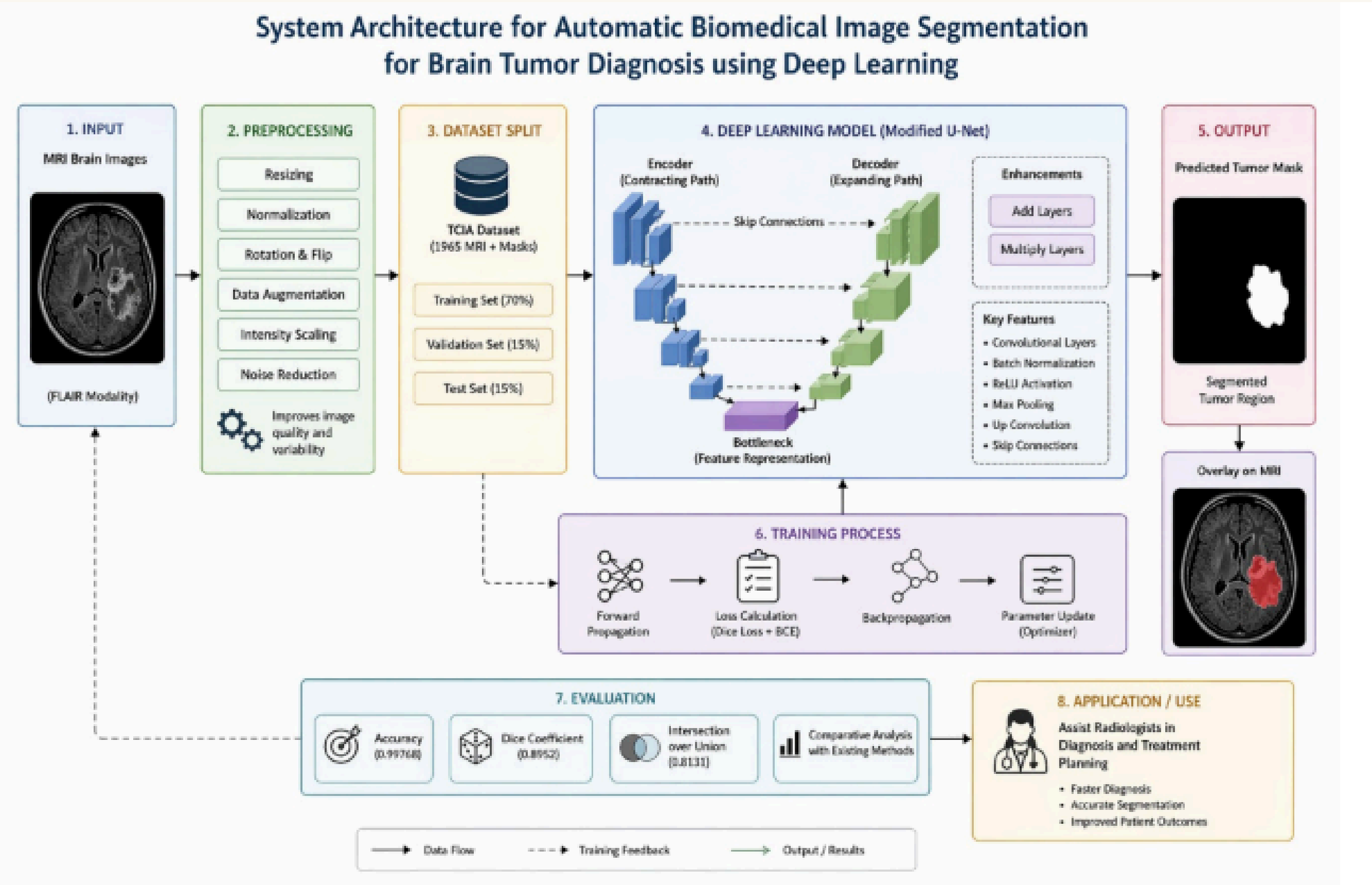
Methodology



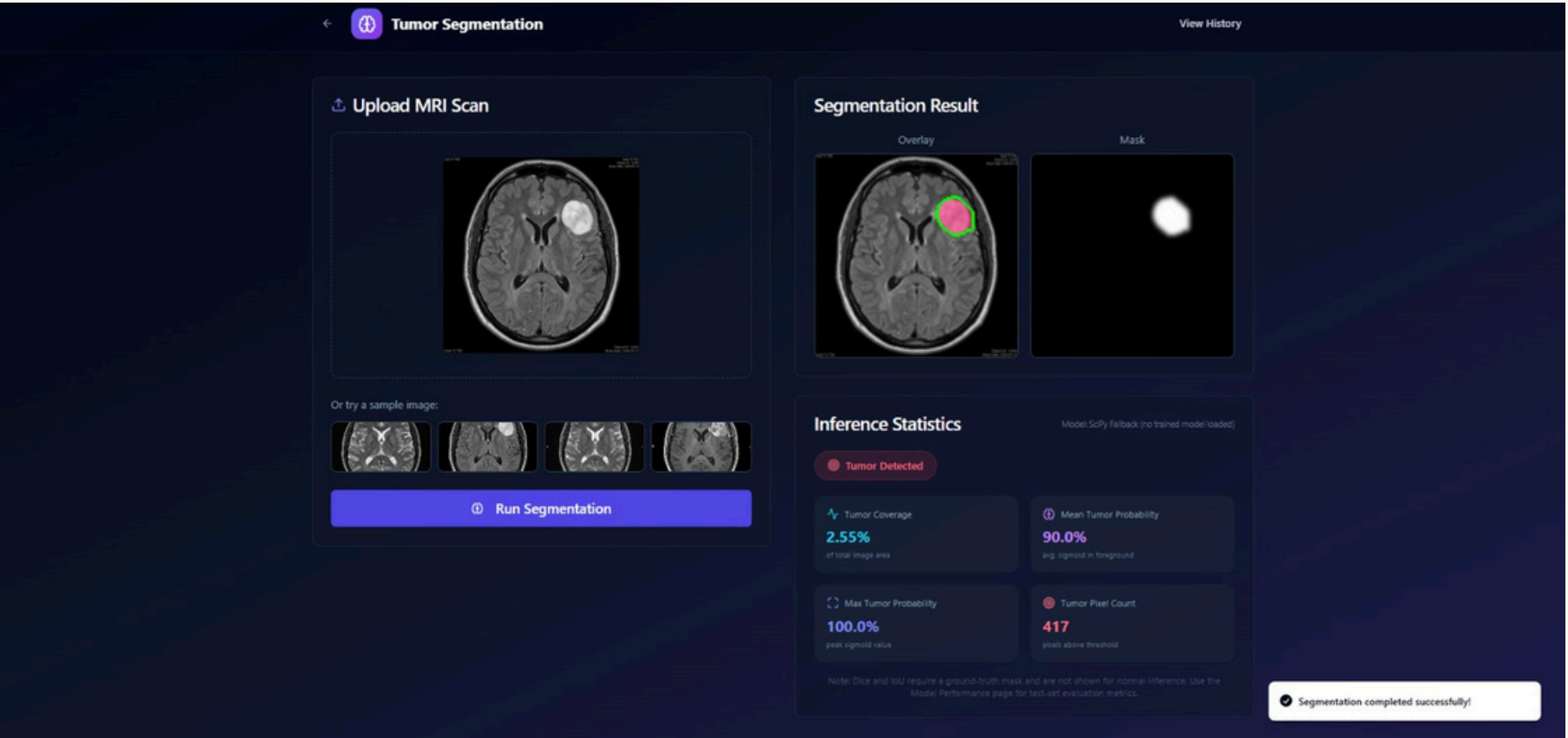
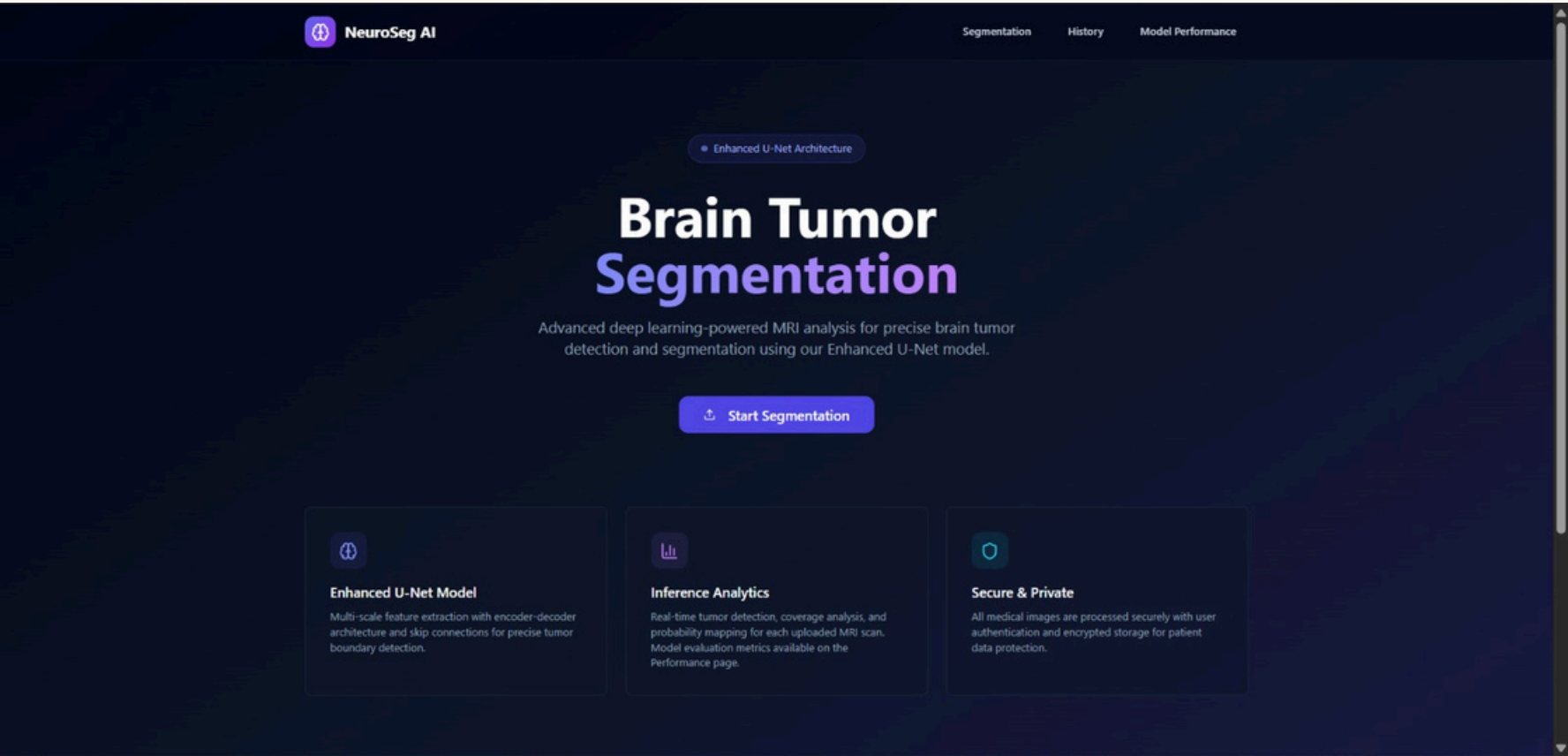
Technology Stack



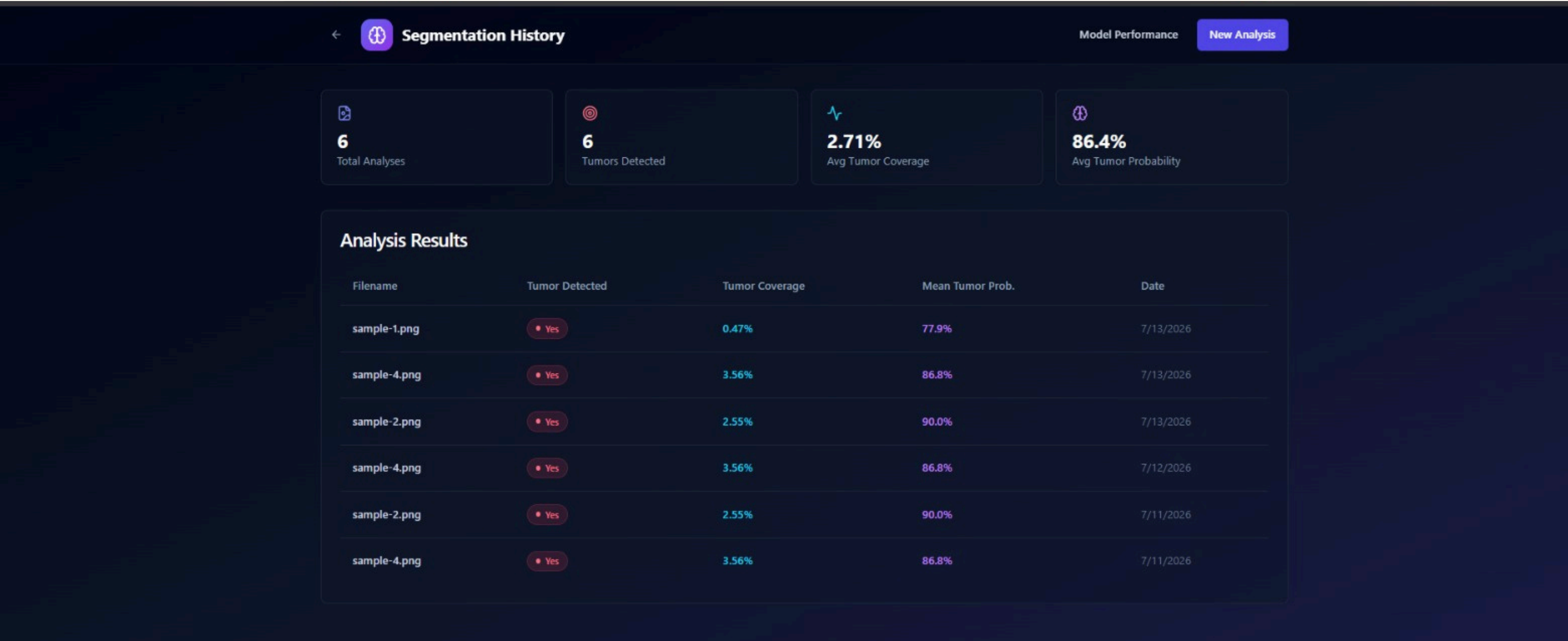
System Architecture



Landing pages:



RESULTS



Existing Results

Achieved 83.65% segmentation accuracy.
Performs basic tumor segmentation.
Limited result tracking.

Our Results

Improved tumor detection and segmentation.
Displays tumor coverage and probability.
Stores complete analysis history.
Provides a web-based MRI analysis system.

Scope

Project Scope

- Develop an AI-based system for automatic brain tumor segmentation.
- Process MRI brain images using deep learning techniques.
- Assist radiologists in identifying tumor regions accurately.
- Provide segmented tumor images for better diagnosis.
- Improve the speed and reliability of tumor detection.

Applications

- Hospitals and healthcare centers.
- Diagnostic laboratories.
- Clinical decision support for radiologists.
- Medical imaging research.
- Brain tumor treatment planning.

Key Features

- Automated MRI image preprocessing.
- Enhanced Modified U-Net architecture for segmentation.
- Accurate tumor localization and visualization.
- High segmentation performance using CNN models.
- Performance evaluation using Accuracy, Dice Score, and IoU.

Future Scope

- Integrate with hospital PACS and healthcare systems.
- Support real-time MRI image analysis.
- Extend to detect multiple neurological disorders.
- Deploy as a cloud-based diagnostic platform.
- Improve accuracy using advanced AI architectures.

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Thank you